VELARIS — Technical Specification Document

Contemporary Streetwear & Sustainable Capsule Wardrobe E-Commerce Platform

Course: E-Commerce (BSc Computer Science)
Stage: Presentation 2 (Week 7 Milestone) — Frozen Specification
Repository: `https://github.com/Denica2208/VELARIS`

Team Members:
• Predut Andreea-Denisa (Frontend & UX Lead)
• Sapca Roman-Stefan (Backend & Database Lead)
Catalogue Scope: 32 Clothing SKUs across 4 Apparel Categories

1. Mandatory Requirements Strategy (A1–A12)

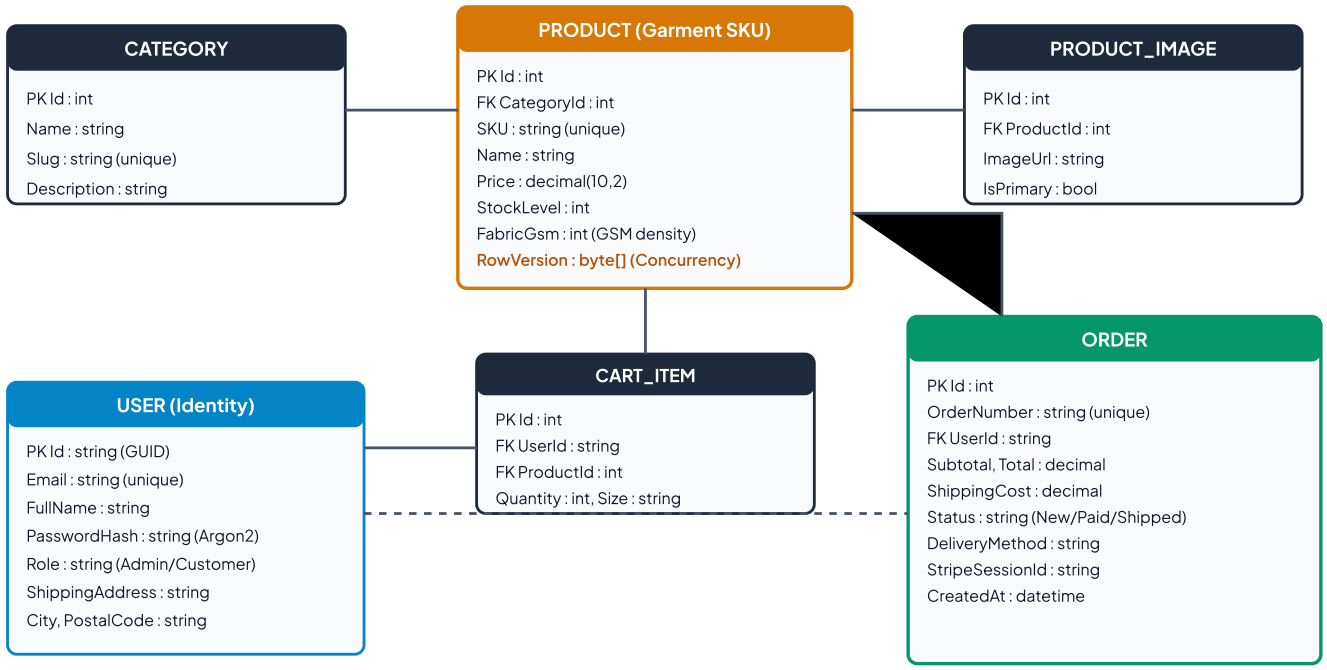
Code	Requirement	Implementation Strategy (1–2 Sentences)	Owner
`A1`	**Product Catalog**	Server-rendered paginated grid capped at 12 items (maximum 24 allowed), dedicated category views, and responsive detail views with fabric GSM density and image galleries.	Predut A.-D.
`A2`	**Basic Search**	Text query execution on product name, fabric composition, and tags using EF Core parameterized SQL queries with distinct empty-state messaging.	Predut A.-D.
`A3`	**Shopping Cart**	Cart items managed client-side with full server-side recalculation and stock validation; authenticated user carts persist across logins in the SQLite database.	Predut A.-D.
`A4`	**User Accounts**	Registration and login secured via ASP.NET Core Identity with password hashing (PBKDF2/Argon2); profiles store customer addresses and order references.	Sapca R.-S.
`A5`	**Checkout Flow**	Four-step wizard: shipping address collection, two distinct carrier options (Standard 15 RON vs. Express 30 RON), order summary review, and Stripe redirect.	Predut A.-D.
`A6`	**Sandbox Payment**	Integration with Stripe Test Mode via official SDK; orders transition to *Paid* exclusively via cryptographically signed webhook confirmation.	Sapca R.-S.
`A7`	**Stock Management**	Atomic stock reduction executed inside a database transaction upon webhook receipt; optimistic concurrency tokens (`RowVersion`) prevent overselling.	Sapca R.-S.
`A8`	**Order History**	Customer portal listing all personal orders with date, ordered garment lines, unit prices, total cost, and status badges (*New, Paid, Shipped, Delivered*).	Predut A.-D.
`A9`	**Administration Panel**	Custom back-office restricted to *Admin* role for CRUD operations on garments/categories with image upload, plus order status lifecycle management.	Sapca R.-S.
`A10`	**Basic Security**	All input validated server-side, automatic anti-CSRF token verification on all POST forms, EF Core ORM parameterized queries, and strict role guards on `/Admin/*` .	Sapca R.-S.
`A11`	**Responsive Design**	Mobile-first Tailwind CSS architecture verified down to 360px viewport width (tested across iPhone SE, Android 360px, iPad, and 1080p desktop).	Predut A.-D.
`A12`	**Automated Tests**	Over 10 xUnit integration tests covering cart calculations, stock deduction limits, and webhook payment confirmation, executable via `dotnet test` .	Sapca R.-S.

2. Chosen Bonus Features

Code	Bonus Feature	Points	Technical Implementation	Owner
`B1`	**Filtered Search**	5 pts	Combinable URL query parameters (`category` , `minPrice` , `maxPrice` , `inStock`) preserved during navigation and sorting.	Predut A.-D.
`B5`	**Discount Codes**	3 pts	Server-validated coupon codes with expiration date, usage quotas, and minimum order values applied during checkout summary.	Sapca R.-S.
`B7`	**Customer Wishlist**	2 pts	Persistent authenticated user favorites list with instantaneous one-click "Move to Cart" item transfer.	Predut A.-D.

3. Entity-Relationship Data Model (ERD)

The relational data model is designed to support high transactional integrity, optimistic concurrency control for garment stock, and authenticated user persistence using SQLite with Entity Framework Core.



Data Schema Highlights & Consistency Rules

- **Atomic Concurrency Guarantee (A7):** The `Product` entity includes an EF Core concurrency token (`RowVersion`). When parallel checkout webhooks confirm payments, if stock drops below requested quantity, the transaction rolls back cleanly with HTTP 409 Conflict.
- **User Persistence & Identity (A4):** Password hashing utilizes ASP.NET Core PBKDF2/Argon2. Saved shipping addresses are linked directly to user accounts, pre-filling the checkout wizard.
- **Auditability & Order Integrity (A8):** `OrderItem` snapshots the exact historical unit price, garment name, and size at purchase time, ensuring product edits do not mutate past invoices.

4. Database Data Dictionary & Integrity Constraints

Detailed schema specifications for primary entities enforcing foreign keys, unique indexes, and domain constraints.

Table: Products (Catalog SKUs)

Column	Data Type	Nullable	Key / Index	Description & Business Rules
`Id`	INTEGER	NO	PK, Auto	Unique internal garment identifier.
`CategoryId`	INTEGER	NO	FK, Indexed	References `Categories(Id)` on delete restrict.
`SKU`	TEXT(32)	NO	UNIQUE	Stock Keeping Unit code (e.g. `VEL-HOD-01`).
`Name`	TEXT(128)	NO	Indexed	Display name used in catalog and search queries.
`Price`	NUMERIC(10,2)	NO	Check > 0	Retail price in RON (recalculated on server).
`StockLevel`	INTEGER	NO	Check >= 0	Available inventory. Decremented on confirmed webhook payment.
`FabricGsm`	INTEGER	NO	Check > 100	Fabric weight in grams per square meter (e.g. 450 GSM).
`RowVersion`	BLOB	NO	Concurrency	EF Core optimistic lock token for race-condition prevention.

Table: Orders & Order Items

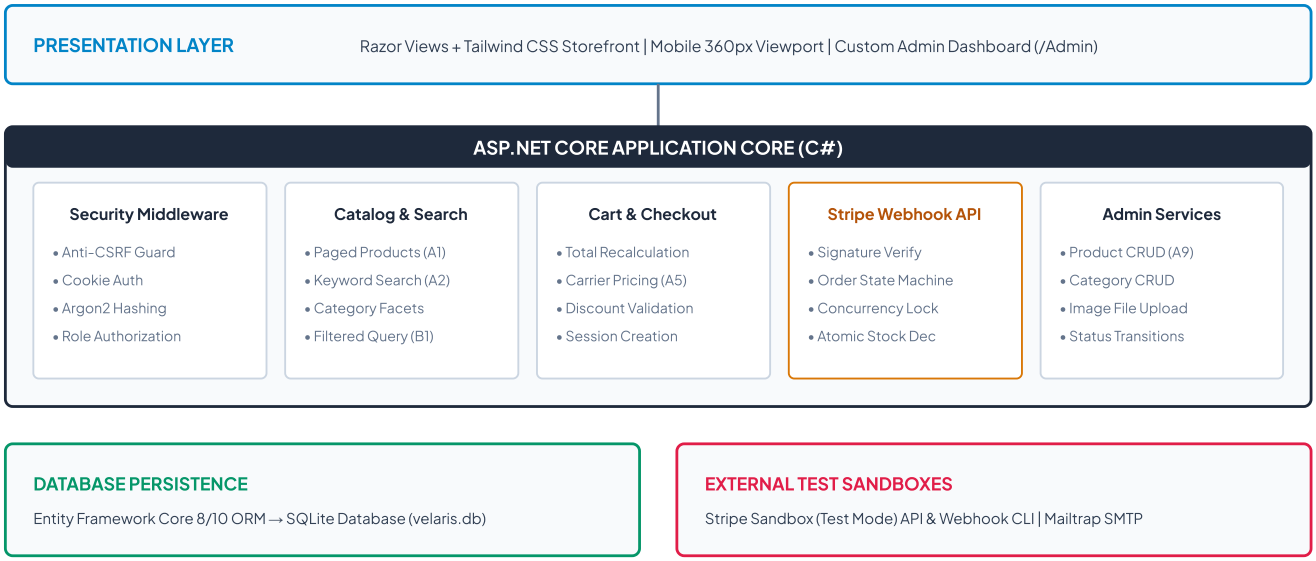
Column	Data Type	Nullable	Key / Index	Description & Business Rules
`Id`	INTEGER	NO	PK, Auto	Internal unique order identifier.
`OrderNumber`	TEXT(32)	NO	UNIQUE	Customer-facing invoice code (e.g. `VEL-2026-8941`).
`UserId`	TEXT(128)	NO	FK, Indexed	References `AspNetUsers(Id)`.
`Status`	TEXT(20)	NO	Indexed	Workflow states: *New* → *Paid* → *Shipped* → *Delivered* / *Cancelled*.
`TotalAmount`	NUMERIC(10,2)	NO	Check > 0	Final captured amount: Subtotal + Shipping – Discount.
`StripeSessionId`	TEXT(128)	YES	UNIQUE	Stripe Checkout session ID used for webhook matching.

Table: DiscountCoupons & Wishlist

Column	Data Type	Nullable	Key / Index	Description & Business Rules
`Code`	TEXT(32)	NO	UNIQUE	Uppercase coupon code (e.g. `VELARIS10`, `CAPSULE20`).
`DiscountPercent`	NUMERIC(5,2)	YES	0–100	Percentage deduction applied on checkout subtotal.
`MaxUses / TimesUsed`	INTEGER	NO	Check >= 0	Enforces quota exhaustion verification on the backend.
`WishlistItems`	COMPOSITE	NO	UNIQUE(U, P)	Pairwise uniqueness of User and Product in saved wishlist.

5. System Architecture & Component Diagram

The platform follows a layered, monolithic architectural model built on ASP.NET Core. This guarantees sub-millisecond local database queries, zero frontend decoupled build latency, and strict server-side validation.



Architectural Principles

- **Separation of Concerns:** Business domain logic is decoupled from HTTP presentation controllers. EF Core repositories enforce data access boundaries.
- **Strict Role Guards (A10):** The admin controller utilizes `[Authorize(Roles = "Admin")]`. Regular customers receive an immediate 403 Forbidden at the routing pipeline level before entering actions.
- **Portability & Single-Command Startup:** Zero external database installation is required. Running `dotnet run` automatically migrates SQLite and seeds the 32 garments.

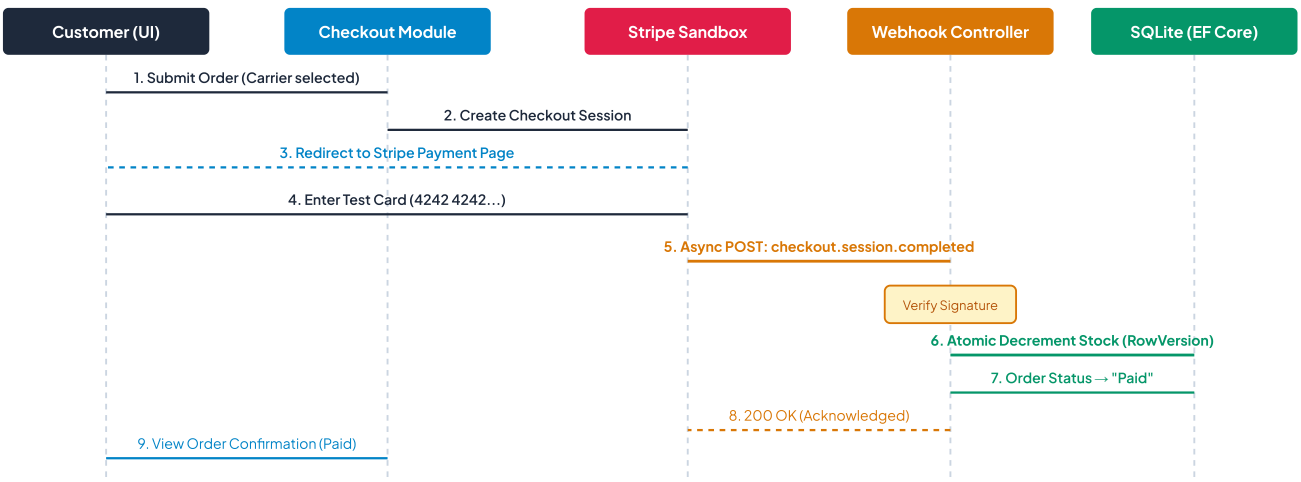
6. RESTful API Endpoints Specification

Comprehensive endpoint mapping including HTTP method, route, authorization role requirement, and expected behavior.

Method	Route	Access Level	Purpose & Payload Summary
POST	/api/auth/register	Public	Register new customer account with Argon2 password hashing. Body: { email, password, fullName, address }.
POST	/api/auth/login	Public	Authenticate credentials; issues secure HTTP-only cookie. Body: { email, password }.
POST	/api/auth/logout	Auth User	Terminates user session cookie and redirects to storefront.
GET	/api/auth/me	Auth User	Returns current authenticated user profile, saved delivery address, and role.
GET	/api/catalog/products	Public	Paginated list of garments (A1). Supports ?page=1&category=hoodies&search=term&sortBy=price-asc.
GET	/api/catalog/products/{slug}	Public	Detailed product information including GSM fabric weight, size availability, and image gallery.
GET	/api/catalog/categories	Public	Lists all 4 active apparel categories with product count totals.
GET	/api/cart	Public / User	Fetches user cart items with live recalculation of subtotal and stock limits (A3).
POST	/api/cart/items	Public / User	Adds garment SKU to cart. Body: { productId, quantity, selectedSize }. Rejects quantity > stock.
DELETE	/api/cart/items/{id}	Public / User	Removes selected item line from cart.
POST	/api/checkout/create-session	Auth User	Validates cart and creates Stripe Checkout session; returns Stripe redirect URL (A5).
POST	/api/stripe/webhook	Stripe Only	Listens for checkout.session.completed; atomically decrements inventory and sets Order Status to *Paid* (A6, A7).
GET	/api/orders/my-orders	Auth User	Lists historical orders for authenticated user with status and item details (A8).
GET	/api/admin/orders	Admin Only	Filterable list of all customer orders by status (New, Paid, Shipped, Delivered) (A9).
POST	/api/admin/orders/{id}/status	Admin Only	Transitions order status (e.g. *Paid* → *Shipped*) (A9).
POST	/api/admin/products	Admin Only	Creates new apparel item with multipart image file upload (A9).
PUT	/api/admin/products/{id}	Admin Only	Updates price, stock levels, GSM weight, or description.
DELETE	/api/admin/products/{id}	Admin Only	Deletes product or marks as archived if historical orders exist.

7. Critical Transaction Flow & Concurrency Control

Requirement **A6** & **A7** specify that an order transitions to *Paid* exclusively upon processor confirmation and that stock is decremented without race conditions.



Concurrency & Webhook Reliability Engineering

- **Optimistic Concurrency Control (A7):** In high-demand drops (e.g. last unit of a Boxy Tee), if two webhook events process concurrently, the database uses SQLite transactional isolation combined with EF Core `DbUpdateConcurrencyException`. The transaction that detects a mutated `RowVersion` aborts and flags the order for refund/unpaid status.
- **Idempotency Enforcement (A6):** Webhook events can be delivered multiple times by Stripe. The webhook controller checks `PaymentTransactions` for the unique `StripePaymentIntentId`; if already processed, it immediately acknowledges with `HTTP 200 OK` without deducting stock twice.
- **Signature Authentication:** The incoming payload is validated using the secret key from `.env` (`STRIPE_WEBHOOK_SECRET`) via `EventUtility.ConstructEvent()`. Tampered or unsigned requests return `HTTP 400 Bad Request`.

8. User Interface Wireframes — Storefront Layouts

Wireframe schematics for primary customer-facing views, engineered for full responsiveness down to 360px viewport widths (Requirement **A11**).

Wireframe 1: Product Catalog Grid (A1, A2, B1) Desktop & Mobile 360px

```
+-----------------------------------------------------------+
| [LOGO: VELARIS] Catalog Hoodies Tees Denim (Cart)| +-----
-----------------------------------------------------+ |
Search: [ Enter keyword... (A2) ] [ Sort: Price ↓ (B1) ]|
| Filter: [x] Heavyweight Fleece Price: [150 - 550 RON]|
+-----------------------------------------------------------+
| [ Image 1 ] | [ Image 2 ] | [ Image 3 ]| | Nocturne
Hoodie | Boxy Minimalist Tee | Selvedge | | 450 GSM
Organic | 280 GSM Cotton | 14oz Denim | | 380 RON [In
Stock] | 180 RON [In Stock] | 420 RON | | [ Add to Cart ]
| [ Add to Cart ] | [View] | | | | | +--------------------
-----------------------------------+ | Pagination (A1): ←
Prev [1] 2 3 Next → (Max 24)| +--------------------------
---------------------------+
```

Features: Responsive 1-col on 360px, 3-col on desktop. Filters preserved in URL query parameters. Empty search displays a clear contextual message.

Wireframe 2: Product Detail Page (A1, B7) SKU Specification

```
+-----------------------------------------------------------+
| ← Back to Hoodies & Sweats | +-------------------------
-+---------------------------+ | | NOCTURNE OVERSIZED
HOODIE | | +-------------------+ | SKU: VEL-HOD-01 | | | |
| 380.00 RON (TVA inclus) | | | Primary Garment | | [ 450
GSM Heavyweight ] | | | Photo (High-Res)| | | | | | |
Select Size: | | +-------------------+ | [ S ] [ M*] [ L ]
[ XL ]| | [Thumb 1] [Thumb 2] | Available Stock: 24 units
| | | | | Fabric & Sustainability: | [ ADD TO CART (A3) ]
| | • 100% GOTS Organic Cotton| [ ♥ Save to Wishlist
(B7)]| | • Pre-shrunk French Terry | | +------------------
---------+---------------------------+
```

Features: Multi-image viewer, real-time stock indicator per size variant, fabric weight specs, and instant wishlist addition.

Wireframe 3: Persistent Shopping Cart (A3, B5) Off-Canvas / Dedicated View

```
+-------------------------------------------------------------------------------------------------+ | YOUR SHOPPING CART (2
items) [ Close × ] | +-----------------------------------------------------------------------------------------------+ |
Garment Item Size Unit Price Quantity Line Total Action | +-----------------------------------------------------------------
-------------------------------+ | Nocturne Oversized Hoodie (450 GSM) M 380.00 RON [- 1 +] 380.00 RON [Delete| | Boxy
Heavyweight Tee - Washed Black L 180.00 RON [- 2 +] 360.00 RON [Delete| +-----------------------------------------------
-----------------------------------------+ | Promo Code (B5): [ VELARIS10 ] [ Apply ] → Discount applied: -10% (-74.00
RON) | | Subtotal: 740.00 RON | Discount: -74.00 RON | Estimated Total: 666.00 RON (Server Computed)| | [ PROCEED TO
CHECKOUT (A5) → ] | +-------------------------------------------------------------------------------------------------+
```

Features: Server-side recalculated subtotals, instant rejection of quantities exceeding stock, authenticated user cart persistence across logouts.

9. User Interface Wireframes — Checkout & Administration

Wireframe 4: Multi-Step Checkout (A5, A6) Wizard Sequence

```
+-----------------------------------------------------+
| CHECKOUT: Step 1 (Address) → Step 2 (Carrier) → Pay | +-
-----------------------------------------------------+ |
Shipping Address (A4 Pre-filled): | | Full Name: [ Andreea
Predut ] | | Address: [ Str. Libertatii nr. 10, Craiova ]
| +-----------------------------------------------------
-+ | Delivery Options (A5 - at least 2 carriers): | | (o)
Fan Courier Standard (2-3 business days) - 15 RON | | ( )
DPD Express Priority (Next day delivery) - 30 RON | +-----
-----------------------------------------------+ |
Order Summary: Items 666 RON + Courier 15 RON = 681 RON| |
[ PAY WITH STRIPE SANDBOX (A6) → ] | +--------------------
-----------------------------------+
```

Wireframe 5: Custom Admin Panel (A9, A10) Role Guarded: Admin

```
+-----------------------------------------------------+
| VELARIS ADMIN BACK-OFFICE [ Logged: admin ] | | [
Products (32) ] [ Categories (4) ] [ Orders (14) ] | +----
-----------------------------------------------------+ |
Orders Filter: [ Status: Paid ↓ ] | | #VEL-2026-8941 |
Andreea Predut | 681 RON | [ Shipped ]| | #VEL-2026-8940 |
Roman Sapca | 380 RON | [ Mark Paid| +--------------------
-----------------------------------+ | Product CRUD
Action: | | [ + Add Garment SKU ] Upload Image: [ Choose
file... ]| | Fields: Name, Category, GSM, Price, Initial
Stock Level| +-----------------------------------------
------------+
```

10. Working Prototype Verification & Seed Data

In accordance with Stage P2 assessment guidelines, a functional prototype is demonstrated live, connecting directly to SQLite and validating the catalogue and authentication flows.

Component	Scope / Volume	Verification Method at P2 Presentation
Seed Catalogue (A1)	32 Garments / 4 Categories	Database populated via EF Core seed script. Live browser pagination, category tabs, and product detail inspection with fabric GSM specs.
Search Engine (A2)	Live text matching	Live query execution against SQLite database; demonstration of populated results and custom zero-results feedback message.
Authentication (A4, A10)	Identity system	Live registration of a new customer account, session login, password verification, profile view, and demonstration that customers are blocked from `/Admin`.
Startup Command	Single command	`dotnet run --project src/Velaris.Store` → Launches on `http://localhost:5000` with zero external dependencies.

P2 Frozen Specification Status: Complete & Approved
Git Milestone Tag: `P2`

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